MAGE: Multisource Attention Network With Discriminative Graph and Informative Entities for Classification of Hyperspectral and LiDAR Data

Di Xiu, Zongxu Pan, *Member, IEEE*, Yirong Wu, and Yuxin Hu

***Abstract*— Land use and land cover (LULC) classification plays a significant role in Earth observation tasks. Nowadays, we can observe the same scene with multiple heterogeneous sensors. Combining diverse information therein for multisource joint classification has become a promising research topic in the remote sensing community. For example, the fusion of hyperspectral image (HSI) and lidar detection and ranging (LiDAR) data has been under active research. The current methodology for HSI and LiDAR joint classification tends to ignore the topological relationship between pixels, limiting the effectiveness of feature extraction and fusion. Another obstacle to satisfactory performance is the scarcity of annotated data. To overcome the above challenges, this article proposes a multisource attention network called multisource attention network with discriminative graph and informative entities (MAGE) to improve the collective classification. We use a semi supervised graph transductive module to underline the relevance among pixels by explicitly constructing a multimodal adjacency matrix. Specifically, MAGE designs a self-supervised feature extraction module for pretraining, mitigating the dependence on annotated samples and alleviating the common overfitting and over-smoothing problems encountered by the deep graph neural network (GNN). The experimental results of three standard datasets, i.e., MUUFL, Trento, and Houston, demonstrate the effectiveness of the proposed approach. In particular, MAGE achieves an overall accuracy of 95.26% and an average accuracy of 96.27% on the challenging MUUFL dataset, surpassing the state-of-the-art (SOTA) methods. The code and models are publicly available at https://github.com/d1x1u/MAGE.**

***Index Terms*— Attention mechanism, contrastive learning, graph neural network (GNN), land use and land cover (LULC) classification, multisource remote sensing.**

I. Introduction

With increasing diversity of data acquired, multisource land use and land cover (LULC) classification has gradually become a focus of research. The various sensors offer unique advantages, so combining the redundant and complementary information from heterogeneous sensors to eliminate the uncertainty associated with a single source and obtain more accurate classification results is expected [1]. For example, although the signals recorded by low-resolution hyperspectral sensors from certain pixels under a complex background suffer from a mixture of substances [2], [3], [4], [5], high-resolution hyperspectral imagery has continuous and narrow bands to capture subtle spectrum differences. In addition, lidar detection and ranging (LiDAR) data provide information on the target altitude even under adverse weather conditions. The efficient combination of the above two sources will provide a more complete view to accurately distinguish objects with similar spectra or heights [6]. However, the heterogeneous data fusion procedure is not trivial for size, resolution, and structural differences. As a result, overcoming domain differences, dealing with redundant information, and combining the benefits of different sensors has become a challenge to the multisource LULC classification task.

Data fusion can be categorized into three main types [7], [8], i.e., observation-level, feature-level, and decision-level fusion, where the observation-level fusion means that raw data are combined directly, and decision-level fusion can be seen as classifier integration. Unlike the other two pre/postprocessing techniques, the feature level is concerned with more effective feature extraction and fusion.

Alongside the emergence of numerous novel convolutional kernels, researchers gradually shift their focus to model engineering instead of feature engineering. Convolutional neural networks (CNNs) can automatically extract deep abstract features, which leads to their widespread use as the feature extractor. As an exemplary for hyperspectral image (HSI) and LiDAR joint classification, [9] proposed a framework consisting of two independent and specialized branches, where a two-tunnel CNN was developed to extract the spectral–spatial features from HSI, and another CNN with the cascade block was used for feature extraction from the LiDAR data. However, vanilla CNNs fail to accurately describe the orientation and semantic change information. The above problem motivated the birth of fractional Gabor convolutional network [10], which used Octave convolution layers to perform multisource information fusion and designed fractional Gabor convolutional layers to extract multiscale spatial features, directional texture features, and frequency variation features. Although the

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multibranch network is an intuitive design for multisource information fusion, [11] found that the multistream architecture essentially played the role of model regularizer instead of being a strictly necessary component by constructing an equivalent single-stream network with dynamic grouping convolution.

However, the widely used training paradigm of supervised CNNs suffers from imbalanced data distribution and imposes heavy demands on the size of annotated training samples. Usually, acquiring labels is notoriously expensive and laborious, with many unlabeled samples readily available wasted. Obtaining satisfactory classification results with a small number of labeled training samples has become the Holy Grail. In this context, a series of unsupervised methods came into being. A feature extraction–fusion network for the HSI and LiDAR data was proposed in [12], which built a multimodal graph for feature fusion and used loss functions including the Laplacian loss and t-distributed stochastic neighbor embedding loss to constrain the feature extraction network. In addition, more work based on the encoder–decoder architecture have emerged by minimizing the error between input samples and their reconstruction. [13] fused the multimodal information by enforcing the fused features to reconstruct the multimodal input in turn. Inspired by the image-to-image translation networks mapping an image from one domain to another domain, [14] designed a patch-to-patch CNN, which first mapped source HSI into a hidden subspace via an encoder, and then reversed to reconstruct the source LiDAR data by a decoder. The hidden representations were deemed as fused features. For improving the deficiency that heterogeneous and complementary information was not efficiently explored in unidirectional mapping designs, the same authors further developed an interleaving perception CNN [15] which used a bidirectional autoencoder to reconstruct the HSI and LiDAR data together and Gram matrix matching to maintain the complementary structure within aligned distributions. Although these methods take full advantage of the large amount of unlabeled data, the noisy relationships between samples are not properly taken into account, limiting the upper bound of model performance.

This article proposes a joint classification framework for HSI and LiDAR sources, namely, multisource attention network with discriminative graph and informative entities (MAGE), which inherits the merits of unsupervised learning to reduce reliance on supervisory information. Furthermore, it explores the graph topological structure concealed in the samples and takes the structural noise into consideration. More specifically, after obtaining the registered images, the network performs two computational streams. First, MAGE generates patches centered at each pixel on HSI and LiDAR images separately. The latter calculates spatial attention masks to enhance the corresponding HSI patches, obtaining observation-level fused spectral–spatial patches for specific pixels. The enhanced patches are fed into the self-supervised feature extraction module to learn embeddings for each pixel. Second, MAGE generates pixel-by-pixel similarity matrices for both the sources, according to the source-dependent proximity measurement, and then weighs the relative importance between the two and introduces the spatial distance constraints to attain a more precise adjacency matrix. Finally, the graph attention network (GAT) module takes the embeddings and the adjacency matrix as inputs to perform subsequent semi supervised classification.

The main contributions of this article are as follows.

1) We spotlight on the graph data structure to express the non-Euclidean topological relationship between pixels within the whole image. Furthermore, we introduce the semi supervised graph transductive module to realize feature aggregation via the self-attention mechanism, allowing better fusion efficacy and improving the tolerance for noisy pixels, which is more flexible than regular convolutional kernels and significant for points located on the classification boundaries.
2) The performance of graph neural networks (GNNs) declines progressively as the number of layers increases, known as the overfitting and over-smoothing problems. As a response, we propose to use a self-supervised feature extraction module to extract features of each pixel in a pretraining way. The experimental results demonstrate that such a design alleviates the problems mentioned above and unleashes the GNN's capacity without resorting to deep structures, even in the case of a small training size.
3) We explore using the attention mechanism at the observation level and feature level, allowing the HSI and LiDAR sources to interact fully. Adding channel attention and spatial attention components helps the network distinguish features of interest from those of irrelevance.
4) We achieve competitive performance on three widely used HSI and LiDAR joint classification datasets, i.e., MUUFL, Trento, and Houston. In particular, MAGE achieves new state-of-the-art (SOTA) performance on the challenging MUUFL dataset.

The remainder of this article is organized as follows. Section II introduces the related work. Section III describes the details of the proposed classification framework. Section IV demonstrates the detailed experimental results and analysis, including ablation study, parameter tuning, and comparative experiments. Finally, Section V summarizes with some concluding remarks.

II. Related Work

A. Attention-Based Models for LULC Classification

The attention mechanism [16], [17], [18], [19], [20] enables models to filter information according to importance. Considering not all the bands within HSI are equally informative and predictive and the usage of useless spectral bands may introduce noises and weaken the performance, a spectral attention component was designed in [21] to recalibrate HSI features by selectively emphasizing informative bands and suppressing less useful ones. Besides integrating data characteristics, the attention mechanism also enables sufficient interaction between the feature extraction modules for different modalities. [22] used the self-attention mechanism to highlight HSI's spectral features and the cross-attention mechanism to accentuate the HSI's spatial features via a LiDAR-derived

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attention map. In the follow-up work, [23], [24] implemented both the spectral attention component and the spatial attention component to facilitate the interaction between the HSI and LiDAR data by leveraging one source to enhance another source.

B. Contrastive Learning for HSI Clustering

Over the past few years, a wave of research on contrastive learning has emerged in the area of computer vision. With the help of agent tasks such as instance discrimination [25], numerous representative works in this field such as momentum contrast (MOCO) [26] and Sim contrastive learning of visual representations (SimCLR) [27] have achieved competitive or even better performance against the supervised methods on some downstream tasks. Most of them are trained by reducing the distance between embeddings of different augmented views of the same image (positive pairs) and increasing the distance between embeddings of augmented views from different images (negative pairs). If only positive samples participate in the training process, the model may collapse to generate the same embeddings indiscriminately. While negative samples play an important role in avoiding model degradation, the training process will be misdirected when samples with the same semantic information are incorrectly designated as negative instances. Therefore, careful construction of positive and negative samples is critical. With the advancement of contrastive learning, a strategy that does not rely on negative pairs, namely, BYOL [28], obtains results on par or superior to its contrastive counterparts. It is a common practice to take the local patch around a pixel as its representative when dealing with hyperspectral imagery. Note that frameworks based on positive and negative sample pairs regard all other samples within the same batch as negative pairs during the training procedure. Since the patch has a size bigger than one, the patches corresponding to adjoining pixels inevitably overlap in a large portion. Although they probably share the same semantic information, those patches will be identified as negative instances under the setting. In the embedding space, multiple feature vectors belonging to the same category are pushed apart due to misleading information, which affects the performance greatly.

The clustering task shares some insights with the classification problems. It splits the dataset into several clusters based on some specific criteria, such that the intracluster similarity and intercluster variability are as large as possible. Under certain circumstances, the clustering algorithms can be used to generate pseudolabels. [29], [30], [31] explored the application of contrastive learning for clustering tasks of HSI, following the pioneering work of contrastive clustering [32], where the instance-level and cluster-level contrastive learning are, respectively, conducted in the row and column space by maximizing the similarities of positive pairs while minimizing those of negative pairs.

C. GNNs for HSI Classification

The graph is a kind of flexible data structure that consists of vertices and edges, capable of expressing a wide variety of non-Euclidean data. In particular, images with regular grid structures are exceptional cases of graphs. Researchers have tried to extend the convolution operator to the graph structure [33], [34], [35], [36], giving rise to GNNs [37]. Compared with CNNs, GNNs can flexibly capture the relationship between distant pixel pairs. Graph convolutional network (GCN) [38] originates from graph signal processing and can be seen as the first-order approximation of ChebNet [36]. Despite its widespread application for the effectiveness and friendly memory overhead especially under constrained computation resources, it is confronted by performance bottleneck. One of the reasons is that GCN performs the neighborhood aggregation step in a way that the connection strength depends on degrees solely. Thus, all the neighboring nodes of the central one with the same degree will be assigned an identical weight, which leads to sensitivity to structural noise and a poor discrimination power of the model. For addressing the shortcomings of prior methods based on graph convolutions or their approximations, GAT [39] used the self-attention mechanism to specify different weights to different nodes in a neighborhood. However, GAT suffers from the static attention, namely, the ranking of the attention scores is unconditioned on the central node, as pointed in [40]. It is worth mentioning that scalable GNNs are of great interest due to the constraints of spatial and temporal complexity. As a representative, graph sample and aggregate (GraphSAGE) [41] proposed a node sampling strategy to reduce the neighborhood size for being applicable on large graphs. Besides, graph sampling based inductive learning method (GraphSAINT) [42] constructs minibatches by sampling the training graph to alleviate the neighbor explosion problem.

GNNs consist of multiple stackable layers and can conceptually be incorporated into a message-passing framework, where nodes attend over the neighbors' features in each layer to carry out neighborhood aggregation and feature transformation. Meanwhile, the number of accessible neighbors explodes exponentially at scale as the number of stacked layers increases if there are no constraints or sampling strategies applied. As a result, GNNs with more stacked layers allow each node to access more neighboring information. However, due to the vanishing gradient problem [43], most SOTA GNNs are mostly shallow models, with performance exceeding the deeper ones by a large margin. For example, GCN essentially performs the Laplace smoothing of the graph signal, making it prone to overfitting and over-smoothing problems as the number of layers increases [44]. In response to this phenomenon, [45] concludes that the optimal number of layers for neighborhood aggregation and feature transformation is inconsistent.

GNNs have also been extensively studied in the field of HSI classification. [46] combined GCN and CNN to break the performance bottleneck of a single model. Specifically, it proposed a minibatch training approach for applying GCN to large-scale HSI datasets based on GraphSAINT [42]. It is worth mentioning that the application of GNNs is not straightforward. One of the inputs to GNNs, the adjacency matrix, reflects the relationships between data points. Unlike standard GNNs' benchmarks such as protein–protein interactions and citation networks, no off-the-shelf adjacency relationship is

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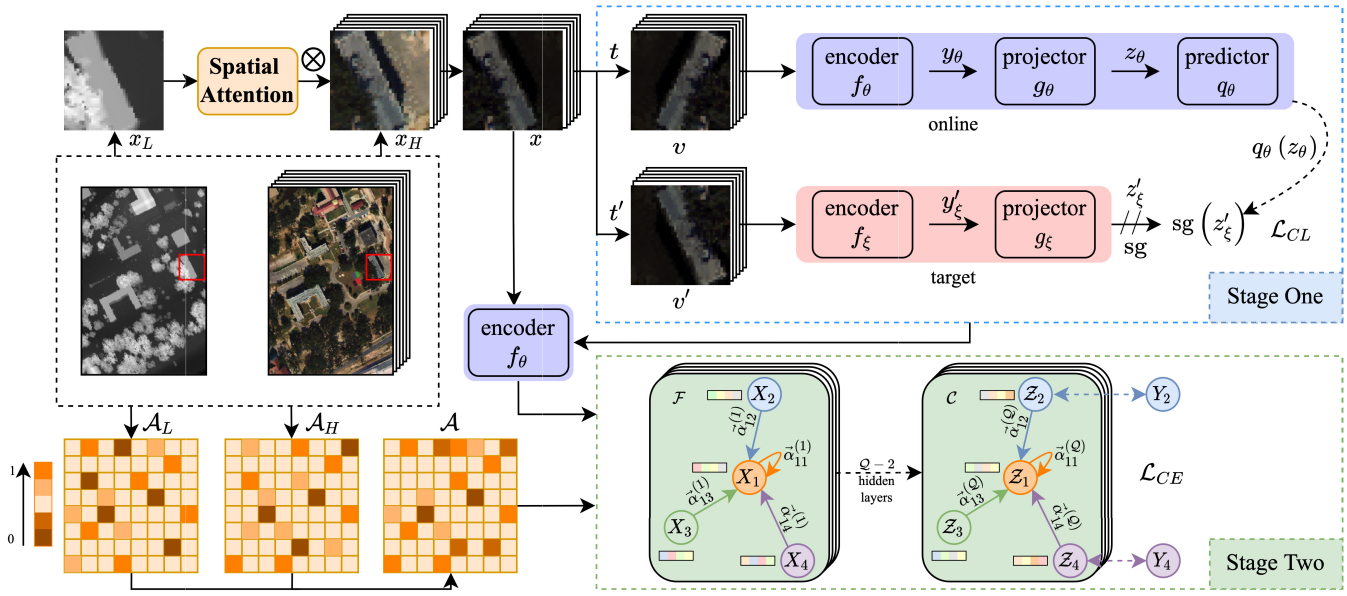

Fig. 1. Proposed multisource classification framework. The network contains two stages, where the upper one is the SFEM, and the lower one is the GAT module consisting of $\mathcal{Q}$ layers for semi supervised classification. As for the similarity matrix in the lower left corner, the similarity levels corresponding to the color blocks gradually increase from dark brown to orange.

available in remote sensing images. To apply GNNs in HSI classification tasks, researchers have come up with diverse ways of constructing adjacency matrices. [47] contended that true labels would often be a subset of pseudolabels and constructed the adjacency matrix in a Semi supervised manner based on the Dirichlet process mixture model, which clustered the data under a Gaussian mixture model assumption and inferred the intrinsic number of mixtures. [48] designed convolutional networks to learn the graph structure in a data-driven way, but the computational overheads were unaffordable.

III. Proposed Classification Framework

In this section, we describe the details for the two stages of MAGE as illustrated in Fig. 1, including the self-supervised feature extraction module (SFEM) for pretraining and semi supervised graph transductive module (SGTM) for classification. In general, SFEM extracts features for each pixel within training set and test set in a self-supervised manner. After that, SGTM constructs a similarity graph and performs semi supervised classification in a transductive way.

Let $\mathcal{H}_{h\times w}^{C_H} \in \mathbb{R}^{h\times w\times C_H}$ and $\mathcal{L}_{h\times w}^{C_L} \in \mathbb{R}^{h\times w\times C_L}$ denote the raw coregistered HSI and LiDAR data, respectively, with C_H and C_L being the channel dimensions, and $N = h \times w$ represents the number of pixels.

A. Self-Supervised Feature Extraction Module

Considering the spatial continuity of HSI, we generate patches $\mathcal{H}^i \in \mathbb{R}^{k\times k\times C_H}$ of size k centered at each pixel $i \in [1, N]$ to obtain the corresponding 3-D data cube. Accordingly, we also generate patches $\mathcal{L}^i \in \mathbb{R}^{k\times k\times C_L}$ at the corresponding position of the LiDAR data.

Our intention is to use the local patch centered on a pixel as its representative. While enriching the features in this way, we hope that the semantic information of the central pixel will be preserved. Due to the complexity of the classification scene, many data points lie on the classification boundary, so some pixels on the patch belong to different classes from the center pixel with relative coordinate $(\lfloor k/2\rfloor, \lfloor k/2\rfloor)$. Before feeding into the feature extractor, those significantly different pixels from the center one should be suppressed, which is achieved by multiplying the mask $\mathcal{M}$ generated by the spatial attention component from the LiDAR patch onto the corresponding HSI patch, whose formula is as 1

$$\mathcal{M}_{\mathrm{mn}}^i = \exp\left(-\frac{\gamma \times \left\|\mathcal{L}_{\mathrm{mn}}^i - \mathcal{L}_{\lfloor k/2\rfloor\lfloor k/2\rfloor}^i\right\|_2}{\sigma^2}\right). \tag{1}$$

The spatial attention component enables the observation-level fusion, where the elevation features in the LiDAR data is exploited in the form of a height filter. After the fusion, the enhanced patch explicitly combines the characteristics from both the sources, including spectral, altimetric, and spatial information. The self-supervised feature extraction module composes of the online network and the target network. The former contains an encoder, a projector, and a predictor, whose weights are denoted by θ, while the latter has the same component composition except for the predictor but different weights ξ. This module takes the enhanced patches to learn features, which are used as the initial node embeddings for the first layer of subsequent GAT module.

In accordance with the fact that patches are characterized by a low spatial resolution and a large number of channels, we design the specific encoder component using the attention mechanism. As illustrated in Fig. 2, the encoder involves three convolutional blocks interspersed with attention components (depicted in Fig. 3). The coordinate attention (CA) component [20] following the first convolution block adaptively highlights the important spatial positions on the feature maps. The residual block [49] emphasizes the informative channels and suppresses the less informative ones through the efficient

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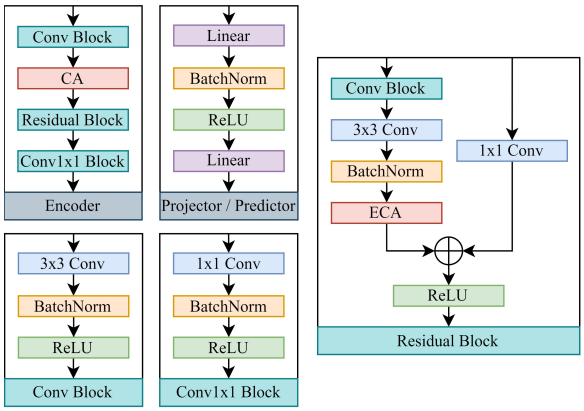

Fig. 2. Schematic of structures of the encoder, projector, and predictor for SFEM.

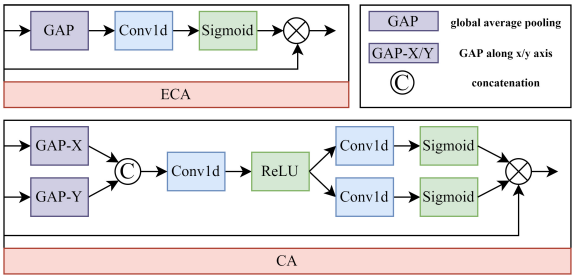

Fig. 3. Schematic of the ECA component and CA component within the encoder.

channel attention (ECA) component [19]. Finally, the last convolutional block integrates the channel information by 1×1 convolution. On the other hand, the projector and predictor share the same configurations, i.e., two multi-layer perceptron (MLP) layers, and the batch normalization (BN) and rectified linear unit (ReLU) layers between them.

The target networks serve as the regression target to train the online networks. During the training process, the stopping gradient operation (depicted as sg in Fig. 1) guarantees that the weights ξ of the target networks are not updated directly by the backpropagation. Instead, they are refreshed as an exponential moving average of the online networks' parameters θ after each training step, i.e., $\xi \leftarrow \tau\xi + (1-\tau)\theta$, where $\tau \in [0, 1]$ represents the decay rate.

Given a set of enhanced patches $\mathcal{D}$, an enhanced patch $x \sim \mathcal{D}$ sampled uniformly from $\mathcal{D}$, and two distributions of image augmentations $\mathcal{T}$ and $\mathcal{T}'$, two augmented views $v \triangleq t(x)$ and $v' \triangleq t'(x)$ from x are produced by applying the corresponding image augmentations $t \sim \mathcal{T}$ and $t' \sim \mathcal{T}'$. Note that the data augmentation pipeline consists of random rotation by integer multiples of 90°, random horizontal or vertical flipping, random crop and resizing, and random erasure of a portion of the area.

The online network calculates the representation $y_\theta \triangleq f_\theta(v)$, the projection $z_\theta \triangleq g_\theta(y_\theta)$, and the prediction $q_\theta(z_\theta)$ from the first augmented view v. The target network generates the representation $y'_\xi \triangleq f_\xi(v')$ and the corresponding projection $z'_\xi \triangleq g_\xi(y'_\xi)$ from the second augmented view v'. Afterward, $q_\theta(z_\theta)$ and z'_ξ are l_2-normalized to $\bar{q}_\theta(z_\theta) \triangleq q_\theta(z_\theta)/\|q_\theta(z_\theta)\|_2$ and $\bar{z}'_\xi \triangleq z'_\xi/\|z'_\xi\|_2$, respectively. The mean squared error between the normalized predictions $\bar{q}_\theta(z_\theta)$ from the online network and normalized projections $\bar{z}'_\xi$ from the target network serves as the objective function

$$\bar{\mathcal{L}}_{\theta,\xi} \triangleq \left\|\bar{q}_\theta(z_\theta) - \bar{z}'_\xi\right\|_2^2 = 2 - 2 \cdot \frac{\left\langle q_\theta(z_\theta), z'_\xi \right\rangle}{\|q_\theta(z_\theta)\|_2 \cdot \left\|z'_\xi\right\|_2}. \quad (2)$$

To make full use of training samples in one forward calculation, we further separately feed v' to the online networks and v to the target networks to compute $\tilde{\mathcal{L}}_{\theta,\xi}$. At each training step, we perform a stochastic optimization step to minimize $\mathcal{L}_{\theta,\xi} = \bar{\mathcal{L}}_{\theta,\xi} + \tilde{\mathcal{L}}_{\theta,\xi}$ with respect to θ. The dynamics are summarized as

$$\theta \leftarrow \text{optimizer}(\theta, \nabla_\theta \mathcal{L}_{\theta,\xi}, \eta) \quad (3)$$

$$\xi \leftarrow \tau\xi + (1-\tau)\theta \quad (4)$$

where η represents the learning rate.

At the end of training, only the encoder f_θ will be kept to extract features of both the training samples and test samples as the initial node embeddings for subsequent module.

B. Semi Supervised Graph Transductive Module

We reshape the raw coregistered HSI $\mathcal{H}_{h\times w}^{C_H}$ and LiDAR data $\mathcal{L}_{h\times w}^{C_L}$ to $\mathcal{H}_N^{C_H} \in \mathbb{R}^{N\times C_H}$ and $\mathcal{L}_N^{C_L} \in \mathbb{R}^{N\times C_L}$, respectively. The reflection vectors in $\mathcal{H}_N^{C_H}$ and $\mathcal{L}_N^{C_L}$ are defined as $h_i = [h_i^1, h_i^2, \ldots, h_i^{C_H}]^\top$ and $l_i = [l_i^1, l_i^2, \ldots, l_i^{C_L}]^\top (i \in [1, N])$, respectively, where $\top$ represents the transpose operation.

A graph portrays the relationships between a collection of entities [50] and is represented by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the vertex set and $\mathcal{E}$ is the edge set. The input adjacency matrix defines the neighborhood relationship. Consistency is the fundamental assumption on the graph, meaning that connected nodes are likely to share the same label, which is the idea behind the classical label propagation (LP) algorithm [51]. In particular, the edges that connect nodes of different classes are considered structural noise. The construction of a sufficiently accurate adjacency matrix is crucial but intractable, because its complexity gradually converges toward the classification problem itself as the noise decreases.

Researchers have made various attempts to reduce the structural noise, including graph structure learning [48], [52], dynamic adjacency matrices [53], [54], etc. However, such approaches suffer from a nonnegligible computation budget. Another stream is the metric-based method, whereas common distance measures face challenges due to intraclass variability, interclass similarity, and the curse of dimensionality [55]. Therefore, we consider the data characteristics of the HSI and LiDAR data to find an appropriate metric, respectively.

Spectral angle mapper (SAM) [56] measures spectral similarity by finding the angle between the spectral signatures h_i and h_j of two pixel vectors

$$\text{SAM}(h_i, h_j) = \cos^{-1}\left(\frac{\langle h_i, h_j \rangle}{\|h_i\|_2 \|h_j\|_2}\right). \quad (5)$$

Unlike the deterministic method SAM, spectral information divergence (SID) [57] is a stochastic measurement method derived from the concept of divergence in information theory,

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which describes the statistics of the spectra by modeling it as a probability distribution.

Let $p_i = (p_i^1, p_i^2, \ldots, p_i^{C_H})$ and $p_j = (p_j^1, p_j^2, \ldots, p_j^{C_H})$ be the normalized probability measure for h_i and h_j, respectively, where $p_*^k = (h_*^k / \sum_{l=1}^{C_H} h_*^l)$. The formulation induces the SID, as follows:

$$\begin{aligned} \mathrm{SID}(h_i, h_j) &= D(h_i \| h_j) + D(h_j \| h_i) \\ D(h_i \| h_j) &= \sum_{l=1}^{C_H} p_i^l \log\left(p_i^l / p_j^l\right) \\ D(h_j \| h_i) &= \sum_{l=1}^{C_H} p_j^l \log\left(p_j^l / p_i^l\right). \end{aligned} \tag{6}$$

A mixed spectral discrimination measure of SID and SAM is developed in [58]. In this article, we take the sinusoidal form as the hyperspectral similarity measure

$$\mathcal{A}_H(h_i, h_j) = \mathrm{SID}(h_i, h_j) \times \sin(\mathrm{SAM}(h_i, h_j)). \tag{7}$$

Usually, the LiDAR reflection vector has a low dimensionality, so we take the Euclidean distance as its similarity measure

$$\mathcal{A}_L(l_i, l_j) = \|l_i - l_j\|_2. \tag{8}$$

Since the HSI and LiDAR data own distinct discriminative abilities, we balance the relative importance of the two sources with a weighting factor λ

$$\mathcal{A}(i, j) = \mathcal{A}_H(h_i, h_j) + \lambda \mathcal{A}_L(l_i, l_j). \tag{9}$$

Considering the local smoothness assumption, i.e., the spatially close points tend to be in identical category [47], [54], we use the logarithm of the Chebyshev distance as an additional constraint, devoting more attention to such prior information, yielding the final joint similarity measure

$$\mathcal{D}(i, j) = \log(d_{i,j}) \times \mathcal{A}(i, j) \tag{10}$$

where $d_{i,j} = \max(|i_h - j_h|, |i_w - j_w|)$, $(\cdot)_h$ and $(\cdot)_w$ represent the vertical and horizontal coordinates of the pixel, respectively.

The adjacency matrix is generated in such a way that for each vertex i, the nodes with the $\mathcal{K}$-nearest distance form its neighborhood $\mathcal{N}(i)$. Formally

$$\mathcal{D}(i, p) \leq \mathcal{D}(i, q) \quad \forall p \in \mathcal{N}(i) \text{ and } q \in \mathcal{S} \backslash \mathcal{N}(i) \tag{11}$$

where $|\mathcal{N}(i)| = \mathcal{K}$, and $\mathcal{S}$ denotes all the labeled nodes.

Note that we adopt the transductive learning setting, namely, all the samples with labels are engaged in constructing the adjacency matrix, including the training samples and testing samples.

Since the structural noise on graphs is inevitable, the network should be resistant to it. We expect to assign more weights to relevant neighbors and less to misleading neighbors by computing the linear combination of neighbors' features with the normalized attention coefficients in the neighborhood aggregation step. The coefficients are calculated via the self-attention mechanism, as 12 shows

$$\begin{aligned} \vec{h}'_v &= \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu} \mathbf{W} \vec{h}_u\right) \\ \alpha_{vu} &= \frac{\vec{a}^T \exp\left(\mathrm{LeakyReLU}\left(\mathbf{W} \cdot \left[\vec{h}_u \bigoplus \vec{h}_v\right]\right)\right)}{\sum_{k \in \mathcal{N}(v)} \vec{a}^T \exp\left(\mathrm{LeakyReLU}\left(\mathbf{W} \cdot \left[\vec{h}_k \bigoplus \vec{h}_v\right]\right)\right)} \end{aligned} \tag{12}$$

where $\vec{h}_v$ and $\vec{h}'_v$ represent the node embeddings before and after the feature transformation, respectively, $\bigoplus$ is the concatenation operation, $\cdot^T$ represents the transposition, and $\mathbf{W}$ and $\vec{a}$ are the learnable weight parameters, applying to every node.

We further extend to the multihead attention mechanisms to stabilize the learning process. $\mathcal{T}$ attention heads execute 12 independently, and then their features are concatenated to obtain the output representation

$$\vec{h}'_v = \bigoplus_{t=1}^{\mathcal{T}} \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(t)} \mathbf{W}^{(t)} \vec{h}_u\right). \tag{13}$$

On the final prediction layer of the network, the averaging operation replaces the concatenation operation to serve as the neighborhood aggregation step. Then, the softmax nonlinear activation function is applied to yield the prediction probability vector for each node

$$\vec{h}'_v = \sigma\left(\frac{1}{\mathcal{T}} \sum_{t=1}^{\mathcal{T}} \sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(t)} \mathbf{W}^{(t)} \vec{h}_u\right). \tag{14}$$

We then calculate the cross-entropy loss over all the labeled samples to optimize trainable parameters as follows:

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{l \in \mathcal{Y}_L} \sum_{f=1}^{C} Y_{\mathrm{lf}} \ln \mathcal{Z}_{\mathrm{lf}} \tag{15}$$

where $\mathcal{Y}_L$, C, Y, and $\mathcal{Z}$ represent the set of labeled nodes, number of classes, true labels, and prediction of probability from the last layer, respectively.

IV. Experiments and Analysis

In this section, three HSI and LiDAR datasets are used to validate the effectiveness of the proposed MAGE. All the programs are implemented using Python 3.8.12. PyTorch 1.10.2 serves as the development framework. The compute unified device architecture (CUDA) version is 11.6. The GNNs are built on PyTorch Geometric [59]. NVIDIA Data Loading Library (DALI) is used to accelerate the data augmentation pipeline. All the experiments are conducted on a server equipped with Ubuntu Server 20.04.4 long term support (LTS), Intel Core i9 10920X, 64-GB RAM, and NVIDIA GeForce RTX 3060. The overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) are used as evaluation metrics. All the experiments are implemented ten times to report the mean and standard deviation. We determine the hyperparameters of the network by random search [60], which can be found at https://github.com/d1x1u/MAGE.

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE MUUFL DATA

Class		Number of Samples	
No.	Name	Train	Test
1	Trees	150	23246
2	Mostly grass	150	4270
3	Mixed ground surface	150	6882
4	Dirt and sand	150	1826
5	Road	150	6687
6	Water	150	466
7	Building Shadow	150	2233
8	Building	150	6240
9	Sidewalk	150	1385
10	Yellow curb	150	183
11	Cloth panels	150	269
Total		1650	53687

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE TRENTO DATA

Class		Number of Samples	
No.	Name	Train	Test
1	Apple trees	129	4034
2	Buildings	125	2903
3	Ground	105	479
4	Woods	154	9123
5	Vineyard	184	10501
6	Roads	122	3174
Total		819	30214

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON DATA

Class		Number of Samples	
No.	Name	Train	Test
1	Health grass	198	1251
2	Stressed grass	190	1254
3	Synthetic grass	192	697
4	Trees	188	1244
5	Soil	186	1242
6	Water	182	325
7	Residential	196	1268
8	Commercial	191	1244
9	Road	193	1252
10	Highway	191	1227
11	Railway	181	1235
12	Parking lot 1	192	1233
13	Parking lot 2	184	469
14	Tennis court	181	428
15	Running track	187	660
Total		2832	15029

A. Data Description

1) MUUFL Data [61]: The scene was collected in November 2010 over the University of Southern Mississippi Gulf Park Campus, located in Long Beach, Mississippi. The dataset contains coregistered HSI and LiDAR-derived digital surface model (DSM). The HSI data are acquired by the ITRES Research Ltd. (ITRES) Compact Airborne Spectrographic Imager (CASI)-1500 sensor, which is composed of 325 $\times$ 220 pixels with 72 spectral channels (64 available bands) ranging from 375 to 1050 nm with 10-nm bandwidth. The spatial resolution of HSI is 0.54 $\times$ 1.0 m, while that of the LiDAR-based DSM is 0.60 $\times$ 0.78 m. The LiDAR data were acquired by the Gemini Airborne Laser Terrain Mapper (ALTM) LiDAR sensor. In this dataset, 11 categories are investigated for the land cover classification task. Table I lists the available training and testing samples.

2) Trento Data [62]: The scene was acquired over a rural area in the south of the city of Trento, Italy. The dataset is composed of the HSI and LiDAR-derived DSM. The HSI data are acquired by the AISA Eagle sensor, which is composed of 600 $\times$ 166 pixels with 63 spectral channels ranging from 0.40 μm to 0.98 μm. The spatial resolution of both the HSI and LiDAR-derived DSM is 1 m. The LiDAR data are acquired by the Optech ALTM 3100 EA sensor. In this dataset, six categories are investigated for the land cover classification task. Table II lists the available number of training and testing samples.

3) Houston Data [63]: The scene was acquired by the National Science Foundation (NSF) Center in June 2012 over the University of Houston campus and neighbor area, Houston Texas, USA. The average height of the sensor above ground was 5500 ft. The dataset is composed of the HSI and LiDAR-derived DSM. The HSI data are acquired by the ITRES CASI-1500 sensor, which is composed of 349 $\times$ 1905 pixels with 144 spectral channels ranging from 364 to 1046 nm at a spectral sampling of 10 nm. The spatial resolution of both the HSI and LiDAR-derived DSM is 2.5 m. In this dataset, 15 categories are investigated for the land cover classification task. Table III lists the available number of training and testing samples.

B. Ablation Study

1) Impact of SFEM: As shown in Table IV, the performance of both the SFEM and SGTM modules versus SGTM module only is evaluated. We find that better performance is achieved consistently across three datasets when using the features learned by SFEM as initial embeddings for the first layer of subsequent SGTM module, compared with training the SGTM module from scratch. On the MUUFL, Trento, and Houston datasets, OA increases by 3.46%, 5.69%, and 5.34%, AA increases by 2.88%, 7.02%, and 4.39%, κ increases by 4.46%, 7.55%, and 5.77%, respectively, after combination with the contrastive learning module. The above experimental results demonstrate the applicability of the features learned by the self-supervised module for the migration of downstream tasks. More specifically, this phenomenon has a potential explanation. According to the message passing mechanism of GNNs, neighborhood aggregation accompanies feature transformation in every layer. However, there exists a contradiction between the two. For too many neighborhood aggregation operations might cause overfitting and over-smoothing problems [44], whereas too few feature transformation operations will reduce the model capacity. So, the self-supervised feature extraction module offers an alternative pretraining solution, mitigating the problems via the decoupled strategy [45]. Such a combination guarantees both the adequacy of feature extraction and the effectiveness of neighborhood aggregation. Table V presents the computational complexity analysis of the SFEM module. The SFEM module accounts for a large proportion of the overall training time.

2) Impact of SGTM: As shown in Table IV, the performance of the SFEM module only versus both the modules is evaluated. The performance under the fine-tuned training strategy is reported for the self-supervised module. Better performance

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TABLE IV

ABLATION ANALYSIS OF THE DESIGNED SFEM AND GAT MODULE IN TERMS OF OA (%), AA (%), AND KAPPA (×100) ON MUUFL, TRENTO, AND HOUSTON DATASETS

Module		MUUFL			Trento			Houston		
SFEM	SGTM	OA	AA	κ	OA	AA	κ	OA	AA	κ
×	✓	91.80±0.22	93.39±0.09	89.33±0.27	93.24±1.68	91.43±1.09	91.02±2.20	89.25±1.28	90.88±1.11	88.38±1.38
✓	×	89.21±0.60	92.40±0.71	86.11±0.77	96.54±1.42	94.47±1.36	95.40±1.86	92.75±0.82	93.38±0.79	92.17±0.88
✓	✓	**95.26±0.10**	**96.27±0.08**	**93.79±0.14**	**98.93±0.35**	**98.45±0.42**	**98.57±0.46**	**94.59±0.28**	**95.27±0.23**	**94.15±0.30**

TABLE V

ELAPSED TRAINING OR TESTING TIME (SECONDS) AND OCCUPIED GRAPHICS PROCESSING UNIT (GPU) MEMORY (GB) FOR THE PROPOSED METHOD ON THREE DATASETS

		MUUFL	Trento	Houston
Training Time	SFEM	2077.48	1085.66	559.03
	SGTM	389.86	139.56	39.32
	Total	2467.34	1225.22	598.35
Testing Time		0.32	0.06	0.05
GPU Memory		7.599	3.391	3.005

TABLE VI

ABLATION ANALYSIS OF ATTENTION COMPONENTS WITHIN THE ENCODERS OF SFEM IN TERMS OF OA (%), AA (%), AND KAPPA (×100) ON THE MUUFL DATASET

CA	ECA	Spat.	OA	AA	κ
×	×	×	90.93±0.51	91.51±0.85	88.16±0.67
✓	×	×	93.82±0.23	94.37±0.10	91.90±0.29
×	✓	×	94.39±0.17	95.62±0.12	92.67±0.21
✓	✓	×	94.86±0.08	96.01±0.07	93.28±0.10
✓	✓	✓	**95.26±0.10**	**96.27±0.08**	**93.79±0.14**

is achieved consistently across three datasets when both the modules are adopted than the self-supervised module itself. On the MUUFL, Trento, and Houston datasets, OA increases by 6.05%, 2.39%, and 1.84%. Following the same trend as OA, AA increases by 3.87%, 3.98%, and 1.89%, respectively. Besides, κ coefficient improves by 7.68%, 3.17%, and 1.98%. The above experimental results demonstrate the full exploitation of the non-Euclidean data structure by GNNs and the effectiveness of the self-attention mechanism in SGTM, where the neighborhood aggregation procedure focuses on neighbors with similar features to the central node, instead of assigning the same weight blindly, sharpening the classification boundary. Table V gives the computational overhead of the SGTM module. There exists unknown pixels outside the training set and testing set, which do not participate in the construction of the graph. On one hand, the ground-truth labels of unknown pixels may be outside the categories to be classified, so excluding them from the construction helps reduce the label noise. On the other hand, the scale of the graph also decreases, making the construction complexity become milder. Upon getting the good initialization features from the SFEM module, the second stage takes less time to reach convergence, being 15.8%, 11.4%, and 6.57% of the overall time on MUUFL, Trento, and Houston, respectively.

3) Impact of attention mechanism: As depicted in Figs. 1–3, the customized spatial attention component of 1 and two other popular attention mechanisms, namely, ECA and CA are adopted. The customized spatial attention component enhances the HSI data with the LiDAR information at the observation level. Concretely, when generating patches centered on a pixel, this component will emphasize pixels similar in height and suppress ones different in height. Such a design specifically helps data points on classification boundaries distinguish ones belonging to different categories. Furthermore, the channel attention component (ECA) strengthens features and reduces redundancy along the channel dimension, and the spatial attention component (CA) pays attention to critical spatial locations. Table VI confirms the validity of the aforementioned attention components. When the spatial attention component is disabled, the self-supervised feature extraction module will merely take HSI patches as input. Compared with not using any attention component, the individual addition of the CA component will improve OA, AA, and κ by 2.89%, 2.86%, and 3.74%, respectively, and the individual addition of the ECA component will improve OA, AA, and κ by 3.46%, 4.11%, and 4.51%, respectively. While the joint use of the CA and ECA modules will improve OA, AA, and κ by 3.93%, 4.50%, and 5.12%, respectively. When the spatial attention component is introduced, OA, AA, and κ will be further improved by 0.40%, 0.26%, and 0.51%, respectively.

4) Impact of similarity measure: As shown in Table VII, we compare the different similarity measure to construct the adjacent matrix of HSI and the final multimodal graph. HSI has hundreds of bands, and such a high-dimensional space incurs the curse of dimensionality. Because of sparsity, distance-based similarity measures such as Euclidean distance have become invalid. Therefore, it is necessary to use a specially designed measure for HSI. The rationality of the above considerations has been experimentally verified. When not using SAM or SID to calculate the similarity between hyperspectral pixels, we use the Euclidean distance to construct the adjacency matrix. Compared with using the Euclidean distance, the individual addition of the SAM component will improve OA, AA, and κ by 0.29%, 0.12%, and 0.38%, respectively, and the individual addition of the SID component will improve OA, AA, and κ by 4.63%, 4.40%, and 5.72%, respectively. While the joint use of SAM and SID will improve OA, AA, and κ by 6.68%, 5.69%, and 8.28%, respectively, and the joint use of SAM and SID will improve OA, AA, and κ by 6.68%, 5.69%, and 8.28%, respectively. Spatial locality is another important constraint, for it alleviates the variabilities of intraclass spatial distribution. When the Chebyshev distance constraint is introduced, OA, AA, and κ will be further increased by 6.69%, 6.39%, and 8.63%, respectively.

C. Parameter Tuning

In this section, we select the MUUFL dataset as benchmark and compute the performance of MAGE with different

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TABLE VII

ABLATION ANALYSIS OF THE HYPERSPECTRAL SIMILARITY MEASURE USED FOR MULTIMODAL GRAPH CONSTRUCTION UNDER THE FRAMEWORK OF MAGE IN TERMS OF OA (%), AA (%), AND KAPPA (×100) ON THE MUUFL DATASET

SAM	SID	Cheb.	OA	AA	κ
×	×	×	81.89±0.69	84.19±0.13	76.88±0.80
✓	×	×	82.18±0.36	84.31±0.31	77.26±0.41
×	✓	×	86.52±0.18	88.59±0.24	82.60±0.21
✓	✓	×	88.57±0.28	89.88±0.17	85.16±0.34
✓	✓	✓	**95.26±0.10**	**96.27±0.08**	**93.79±0.14**

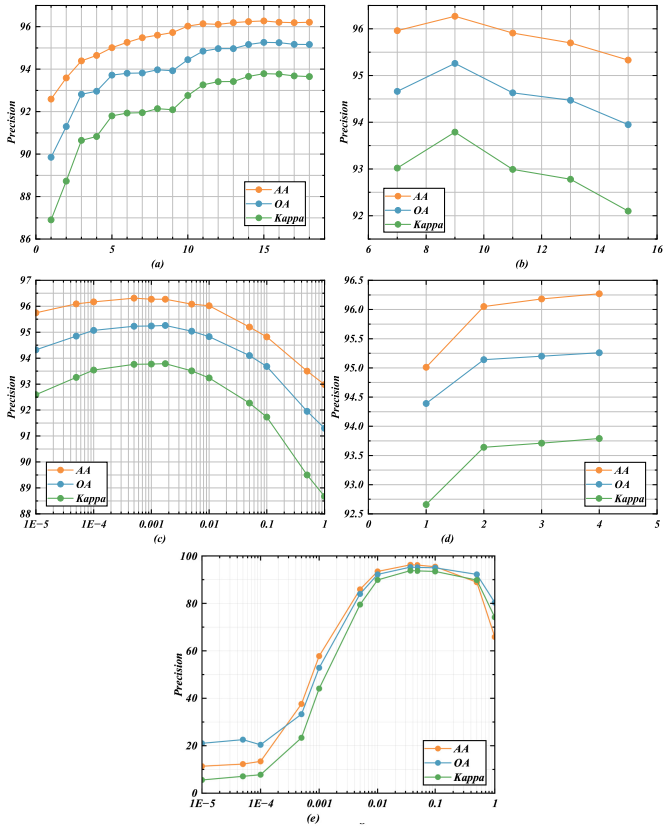

Fig. 4. Parameter robustness analysis on the MUUFL dataset.

hyperparameter values. In addition, the precision under a small size of training samples is tested.

1) The Number of Neighbors: We conduct experiments using MAGE with the number of neighbors from two to 18 and achieve the best result when the number of neighbors is set to 15. The trends can be found in Fig. 4(a). As the number of neighbors increases, more neighbor information can be accessed during the neighborhood aggregation step. Note that the accuracy gradually improves and converges to saturation as the number of neighbors increases. Owing to the attention mechanism, important neighbors should permanently be assigned greater weights, so the model capacity is expected to consistently increase when more neighbors are considered, and the performance should not be considerably inferior to a smaller number of neighbors.

2) The Patch Size: The performance of MAGE with different patch sizes is evaluated in Fig. 4(b), and the best result is achieved when the patch size is set to nine on the MUUFL dataset. Due to the assumption of locality, it can be assumed that a pixel and its neighbors might belong to the same category. Therefore, we generate a corresponding patch to represent the pixel. However, points located on the classification boundary will incorporate many pixels in different categories into the same patch, producing some interference. Although all kinds of attention mechanism within MAGE have taken this problem into account, an oversized patch will reduce the computational accuracy and efficiency. The appropriate patch size closely depends on the resolution of the dataset.

TABLE VIII

CLASSIFICATION PERFORMANCE WITH DIFFERENT SIZES OF TRAINING SAMPLES IN TERMS OF OA (%), AA (%), AND KAPPA (×100) ON THE MUUFL DATASET

	20%	40%	60%	80%	100%
OA	89.77±0.35	92.99±0.38	93.76±0.33	94.06±0.19	**95.26±0.10**
AA	90.80±0.28	93.69±0.21	94.80±0.12	95.70±0.09	**96.27±0.08**
κ	86.68±0.43	90.83±0.48	91.85±0.43	92.25±0.24	**93.79±0.14**

3) The Balance Factor: We then investigate the sensitivity of MAGE to the balance factor λ in Fig. 4(c), and the best result is achieved when λ is taken as 0.00175. As expressed in 9, λ controls the relative importance of the LiDAR similarity measure. The experimental results demonstrate that either too small or too large λ values lead to performance degradation. Small λ causes that only hyperspectral information is used for the final adjacency matrix construction, with the height information lost. One can conclude that effective fusion of multimodal information can bring about superior performance than unimodality. On the other hand, the LiDAR information gradually dominates in the construction of the multimodal adjacency matrix as λ grows larger, with the rich discriminative information in hyperspectral imagery ignored.

4) The Number of Attention Heads: Fig. 4(d) studies the performance of MAGE under different number of attention heads within the SGTM module from one to four being analyzed, and the best result is achieved when it is set to four. When the model expands from single attention head to more attention heads, the performance leaps obviously. Similar to the role of multiple channels in convolutional kernels, additional attention heads enlarge the model capacity by allowing more patterns and enable it to learn different aspects of knowledge under the premise of not causing overfitting.

5) The Learning Rate: Fig. 4(e) displays the effect of learning rate for the SGTM module on the final classification precision. The accuracy keeps improving as the learning rate increases from 0.0001 to 0.1. The performance of MAGE starts to deteriorate when it continues to increase from 0.1. The model converges well when the learning rate falls in the interval between 0.01 and 0.1.

6) Classification Performance With Different Sizes of Training Samples: We have provided the quantitative analysis of the impact regarding the constitution of the training and testing data, as shown in Table VIII, where the setting of 100% corresponds to the default number of training and testing samples for MUUFL in Table I. As for other p% settings, p% of default training samples will be randomly sampled as the training set. Compared with the 20% setting, MAGE trained with the default split will improve the classification accuracy by 6.12%

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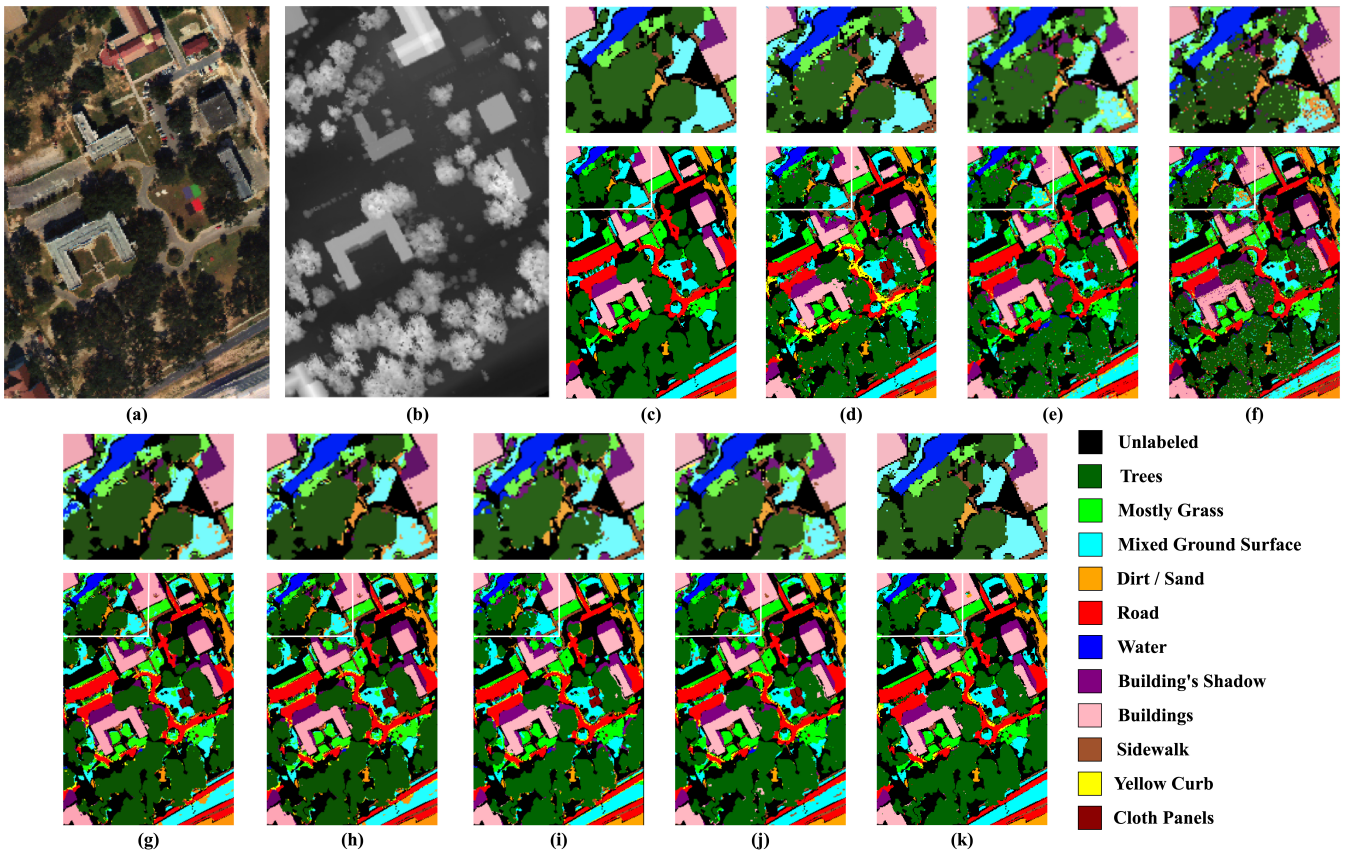

Fig. 5. MUUFL hyperspectral imagery and LiDAR dataset with classification maps. (a) False-color hyperspectral imagery. (b) LiDAR imagery. (c) Ground-truth map. (d) LP. (e) MLP. (f) EndNet. (g) MDL-Middle. (h) MDL-Cross. (i) FusAtNet. (j) S2ENet. (k) MAGE.

relatively (from 89.77% to 95.26%). Benefiting from virtues of the SFEM module, MAGE exhibits a moderate performance degradation under extremely limited training samples, which significantly facilitates the application in real-world scenarios.

D. Comparative Experiments

We conduct comparative experiments against following methods for HSI and LiDAR joint classification. The assessment methods include visual comparisons and quantitative analysis.

1) *LP [51]:* An iterative semi supervised algorithm to propagate labels through the dataset along high-density areas defined by unlabeled data until convergence, where a node's label propagates to all the nodes according to their proximity.
2) *MLP [64]:* MLP consisting of three layers, using dropout to prevent overfitting.
3) *EndNet [13]:* A network following deep encoder–decoder network architecture. It uses the reconstruction strategy for feature fusion.
4) *MDL-Middle [1]:* A concatenation-based fusion framework. It stacks the outputs derived from different streams in networks.
5) *MDL-Cross [1]:* The framework seeks to learn compact feature representations across modalities by interactively updating the parameters.
6) *FusAtNet [22]:* The framework uses HSI modality to highlight its own spectral features via a self-attention mechanism, and a cross-attention approach to harness the LiDAR-derived attention map that accentuates the spatial features of HSI.
7) *S2ENet [23]:* The framework uses a spatial–spectral enhancement module for cross-modal information interaction. Specifically, spatial enhancement module enhances the spatial representation of the hyperspectral data by LiDAR features, and spectral enhancement module enhances the spectral representation of the LiDAR data by hyperspectral features.

Quantitative and qualitative comparison results for the MUUFL dataset are listed in Table IX and Fig. 5. Under the circumstance of a dense distribution of labeled samples and a small size of training data, MAGE obtains an obvious performance gain for its full consideration of topological structure, spatio-spectral features, and adequate cross-modality interaction. In the implementation of MLP, all the individual hyperspectral and LiDAR bands are concatenated to serve as the fusion strategy. The lack of spatial continuity and feature interaction leads to less discriminative features. MAGE exceeds MLP by 9.44% OA, 10.32% AA, and 12.13% Kappa. Notably, LP achieves salient results by making full use of the structural information. However, the model's capacity is limited due to neglect of structural noise and abundant features. LP is inferior to MAGE by 7.31% OA, 3.79% AA,

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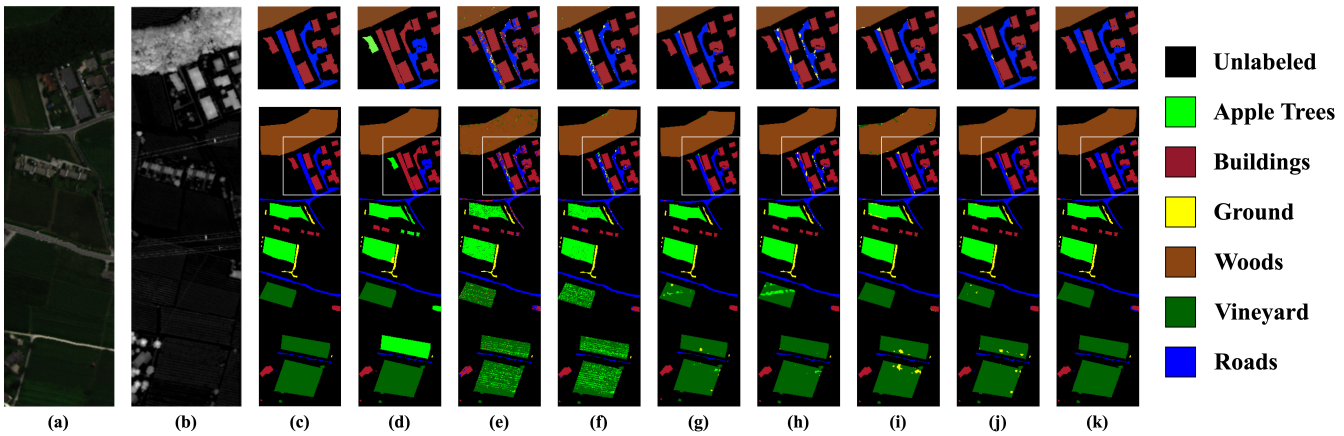

Fig. 6. Trento hyperspectral imagery and LiDAR dataset with classification maps. (a) False-color hyperspectral imagery. (b) LiDAR imagery. (c) Ground-truth map. (d) LP. (e) MLP. (f) EndNet. (g) MDL-Middle. (h) MDL-Cross. (i) FusAtNet. (j) S2ENet. (k) MAGE.

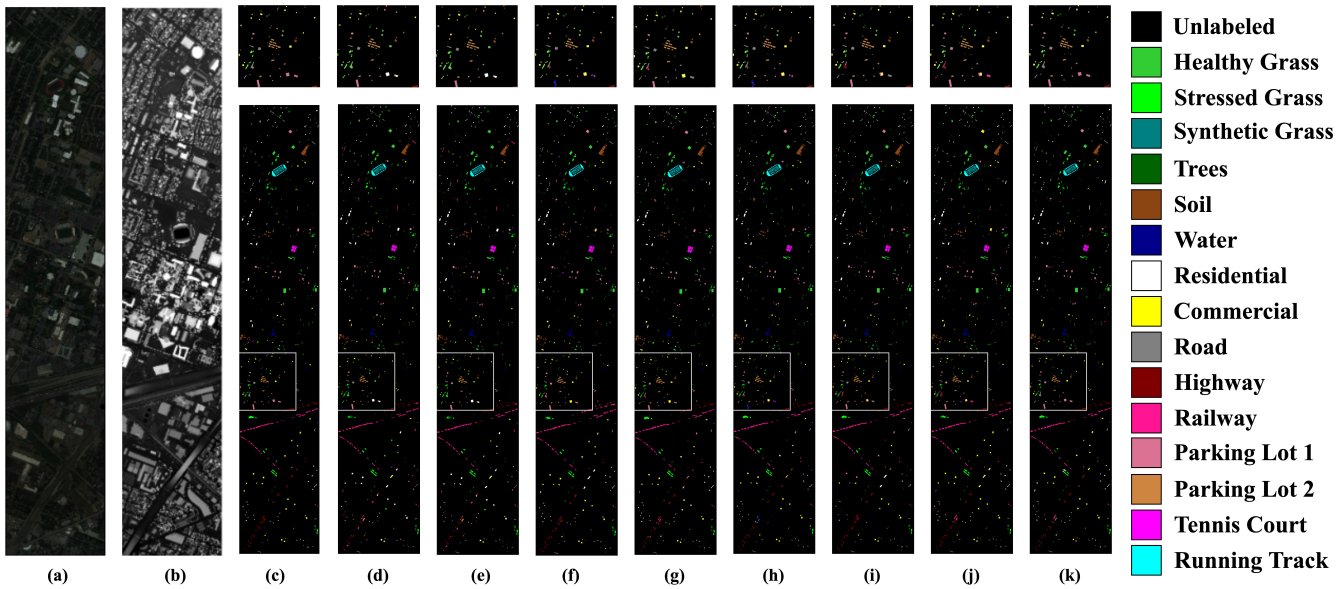

Fig. 7. Houston hyperspectral imagery and LiDAR dataset with classification maps. (a) False-color hyperspectral imagery. (b) LiDAR imagery. (c) Ground truth map. (d) LP. (e) MLP. (f) EndNet. (g) MDL-Middle. (h) MDL-Cross. (i) FusAtNet. (j) S2ENet. (k) MAGE.

TABLE IX

COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE MUUFL DATA

No.	Class	Performance							
		LP	MLP	EndNet	MDL-Middle	MDL-Cross	FusAtNet	S2ENet	MAGE
1	Trees	88.18	89.23±0.54	85.06±1.46	90.25±1.03	86.23±2.81	89.02±2.85	89.70±1.73	96.04±0.21
2	Mostly grass	89.23	80.34±4.36	75.42±10.70	85.67±7.84	79.13±15.30	83.96±6.92	88.96±6.63	95.36±0.35
3	Mixed ground surface	82.88	71.70±4.75	74.21±5.21	77.88±7.15	67.40±13.79	78.45±6.27	70.80±16.23	93.00±0.52
4	Dirt and sand	98.47	77.66±3.88	91.09±4.22	95.73±2.83	75.51±8.90	90.25±7.14	93.50±5.11	97.67±0.24
5	Road	79.08	89.56±1.20	89.89±1.23	89.67±1.77	79.92±7.14	89.77±2.49	87.81±3.90	92.83±0.35
6	Water	100.00	98.95±0.12	98.88±0.43	99.72±0.27	89.31±0.27	99.55±0.38	99.66±0.35	99.76±0.06
7	Building shadow	96.64	90.05±0.54	89.29±3.48	96.22±1.05	95.34±2.82	95.93±2.49	97.04±1.36	97.25±0.33
8	Buildings	92.95	91.51±0.56	91.51±1.20	96.14±0.94	90.81±1.65	96.13±1.65	96.04±1.17	96.05±0.31
9	Sidewalk	89.82	69.11±5.61	72.38±8.40	85.18±4.72	73.57±6.79	87.20±4.28	87.77±5.62	91.98±0.81
10	Yellow curb	100.00	94.21±2.26	98.52±1.71	99.78±0.27	95.30±0.56	94.70±4.87	98.74±2.00	99.07±0.43
11	Cloth panels	100.00	93.09±0.85	99.93±0.15	99.33±0.46	98.25±1.13	99.37±0.29	99.18±0.55	99.93±0.15
	OA	87.95	85.82±0.43	84.55±0.87	89.38±0.79	82.80±2.42	88.63±1.61	88.27±2.23	**95.26±0.10**
	AA	92.48	85.95±0.60	87.83±1.33	92.33±0.75	84.62±2.08	91.30±1.11	91.75±1.37	**96.27±0.08**
	κ	84.50	81.66±0.54	80.20±1.12	86.28±1.00	78.12±2.93	85.37±1.98	84.90±2.80	**93.79±0.14**

and 9.29% Kappa. In terms of visual comparison results, the proposed MAGE attains smoother classification results with less intraclass noises and clearer classification boundaries as shown in the zoomed complicate region covering multiple categories in Fig. 5.

Table X and Fig. 6 present the comparison results of the evaluation criteria and classification maps over the Trento dataset. Samples from different categories tend to be concentrated in separate regions, with the exception that buildings and roads are close in the highlighted area. Spatial homogeneity

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TABLE X
COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE TRENTO DATA

No.	Class	Performance							
		LP	MLP	EndNet	MDL-Middle	MDL-Cross	FusAtNet	S2ENet	MAGE
1	Apple trees	98.31	90.45±1.00	88.17±19.37	99.60±0.58	99.20±1.16	95.26±7.51	99.12±1.49	98.55±1.42
2	Buildings	72.89	89.02±1.98	96.79±0.61	99.72±0.22	99.14±0.54	98.35±1.70	99.82±0.07	95.82±1.37
3	Ground	75.99	97.22±0.42	97.70±0.63	99.67±0.47	99.67±0.38	96.26±6.27	99.94±0.19	100.00±0.00
4	Woods	100.00	96.71±0.71	99.16±0.40	99.87±0.10	99.81±0.24	78.27±38.66	99.69±0.39	99.61±0.45
5	Vineyard	71.97	83.71±1.75	78.86±8.27	97.98±1.19	96.35±3.92	92.77±16.31	99.46±0.46	99.95±0.14
6	Roads	70.01	71.65±2.31	88.94±1.12	90.76±1.54	82.61±16.56	91.52±3.86	91.13±0.86	96.73±1.06
OA		83.90	87.99±0.54	89.31±1.50	98.20±0.52	96.65±2.95	89.18±12.78	98.65±0.23	**98.93±0.35**
AA		81.53	88.13±0.39	91.60±2.15	97.93±0.39	96.13±3.37	92.07±7.50	98.19±0.32	**98.45±0.42**
κ		79.08	84.13±0.69	85.98±1.94	97.61±0.68	95.59±3.82	86.32±15.87	98.20±0.31	**98.57±0.46**

TABLE XI
COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE HOUSTON DATA

No.	Class	Performance							
		LP	MLP	EndNet	MDL-Middle	MDL-Cross	FusAtNet	S2ENet	MAGE
1	Healthy grass	89.05	85.40±0.46	86.47±1.21	85.90±0.36	83.88±2.18	85.80±1.29	86.09±0.45	89.90±1.39
2	Stressed grass	86.44	95.89±0.82	93.98±3.92	99.25±0.87	98.41±3.55	98.80±1.81	99.33±1.58	97.38±1.29
3	Synthetic grass	99.86	99.83±0.06	99.84±0.12	99.96±0.09	99.99±0.04	99.96±0.07	100.00±0.00	100.00±0.00
4	Trees	98.47	99.29±0.24	98.05±0.62	99.68±0.31	99.36±0.45	99.13±0.53	99.27±0.43	98.87±0.44
5	Soil	98.63	96.20±1.57	97.82±1.87	99.98±0.03	99.65±0.30	97.87±4.79	99.99±0.02	99.82±0.03
6	Water	99.08	97.91±0.69	97.97±0.94	99.57±0.57	99.35±0.73	98.89±0.80	98.89±0.91	100.00±0.00
7	Residential	76.58	88.18±1.58	82.89±4.49	93.91±2.02	90.39±3.72	91.42±4.77	90.79±3.66	93.79±1.60
8	Commercial	42.52	45.59±4.08	85.35±4.21	90.44±3.43	94.16±4.42	88.95±6.11	94.00±2.10	90.99±2.55
9	Road	60.30	76.57±2.35	69.37±19.65	89.73±1.89	92.28±4.25	89.32±5.02	85.47±8.08	91.32±2.14
10	Highway	61.86	92.24±4.02	74.50±13.32	76.65±8.87	77.63±9.70	79.75±10.98	87.09±10.36	81.17±3.47
11	Railway	90.93	87.83±1.91	93.70±4.34	96.35±1.50	96.13±1.68	93.06±3.56	97.94±2.02	98.79±0.44
12	Parking Lot 1	57.75	57.75±13.36	78.42±11.85	88.22±4.31	60.52±20.81	85.00±9.19	87.48±4.24	95.91±2.57
13	Parking Lot 2	87.21	58.96±6.36	80.58±12.13	93.84±2.05	90.87±3.80	91.64±2.62	91.30±1.54	91.04±1.69
14	Tennis Court	99.77	99.32±0.47	99.86±0.24	100.00±0.00	99.95±0.09	97.99±4.07	99.98±0.07	100.00±0.00
15	Running Track	100.00	99.02±0.18	98.92±0.43	99.98±0.05	99.52±0.76	99.38±0.56	100.00±0.00	100.00±0.00
OA		79.92	84.12±1.22	87.74±3.03	93.20±0.75	90.80±2.33	92.12±1.58	93.69±1.45	**94.59±0.28**
AA		83.23	85.33±1.21	89.18±2.53	94.23±0.68	92.14±1.90	93.13±1.36	94.51±1.18	**95.27±0.23**
κ		78.30	82.85±1.32	86.76±3.26	92.65±0.81	90.06±2.50	91.48±1.70	93.18±1.57	**94.15±0.30**

reduces the intraclass variability, which facilitates accurate classification. In the prominent enlarged region, it is conceivable that buildings and roads exhibit a large difference in height, so effective integration of LiDAR data will significantly improve the model's capability. Methods without attention mechanism such as multimodal deep learning (MDL)-Middle and MDL-Cross face a great deal of confusion in the junction regions. Aided by the spatial attention components, MAGE provides better results in these difficult areas, with an OA of 0.73% and 2.28% superior to the above methods, respectively.

As depicted in Fig. 7, the Houston dataset is characterized by a huge scale and sparsely distributed labeled samples, which leads to an increase in intraclass difference and interclass similarity. Benefiting from the hypothesis of spatial locality, MAGE is resistant to the changes in distribution to some extent. Within the amplified dense region, MAGE achieves the most consistent classification results with the ground truth in comparison to other methods. Table XI exhibits the numerical precision across different methods. Compared with EndNet, a representative of the unsupervised model, as well as FusAtNet and spatial-spectral cross-modal enhancement network (S2ENet), representatives of the supervised attention networks, MAGE outperforms them by 6.85%, 2.47%, and 0.90% OA, 6.09%, 4.14%, and 0.76% AA, and 7.39%, 2.67%, and 0.97% Kappa, respectively.

V. CONCLUSION

HSIs have prominent spatial–spectral resolution and provide rich discriminative information. Under the assistance of elevation information from the LiDAR data, an effective classification framework is expected to achieve better classification precision. This article proposes a multisource information fusion framework called MAGE for joint land cover and land use classification with the HSI and LiDAR data. MAGE effectively combines the self-supervised feature extraction module and the semi supervised GAT module, taking full advantage of unlabeled samples and the topological relationships among pixels. In addition, we fully explore the use of attention mechanism. Experiments on three benchmark datasets demonstrate the effectiveness of the proposed approach. In particular, MAGE achieves outstanding results on the challenging MUUFL dataset. In the subsequent work, we consider further improving the feature extractor within the self-supervised module to enhance its feature extraction capability. We will also explore methods to improve the robustness against structural noise. Moreover, we will try to extend this method to other multisource remote sensing joint classification datasets to verify its universality. Eventually, with the increase in pixels to be classified, the LULC classification task tends to be equivalent to the semantic segmentation task [65], [66]. Therefore, it would be a promising research direction to explore the application of our model on segmentation tasks.

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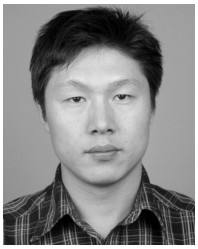

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